

Live System Audit & Diagnostics Report

Direct End-to-End Verification of Application Routing, Live Network Fetching, Data Integrity, and Malfunction Analysis.

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ROUTES OPERATIONAL (200 OK)

Scanner Active

REAL LIVE HTTP/SSL/RDAP

Discovery Mocked

SYNTHETIC SEED POOL

1. Is the Website Fetching Live Data? (Critical Finding)

YES — The Website Scanner (/scanner) Fetches 100% Real Live Data

When a user inputs any domain or batch of URLs (e.g. `vercel.com`, `stripe.com`, `linear.app`), the server executes real-time HTTP requests, checks DNS, inspects security headers, measures TTFB latency, queries the RDAP protocol for domain age, and triggers asynchronous Google Lighthouse PageSpeed audits.

NO — The Lead Discovery Engine (/discovery) & Seed Database Use Synthetic Simulation

The Discovery module and initial 140-lead database are generated by a deterministic generator algorithm (`src/lib/mock/leads.ts`). They do *not* query live Google Search or Google Places APIs yet (scheduled for Phase 2).

2. Root Cause Analysis: Repeating Websites & Broken Links

2.1 Why are Websites Repeating?

In `src/lib/mock/leads.ts` (lines 131–186), the generator uses a hardcoded pool of **35 popular tech domains** (`canva.com`, `github.com`, `stripe.com`, `shadcn.com`, etc.) for all "ONLINE" leads.

- When the seed script or Discovery console generates 140 leads, it cycles through these 35 domains using modulo arithmetic: `domain = REAL_ONLINE_DOMAINS[(i + seed) % 35]`.
- This causes distinct synthetic company names (e.g. *Urban Motors*, *Nova Goods*, *Vertex Supply Co*) to be paired with the same repetitive domains (e.g. `canva.com`).

2.2 Why are Some Websites Broken & Not Openable?

When the generator creates leads with problem types like `BROKEN_SITE`, `NO_SSL`, or `OFFLINE`, it does *not* assign a real domain. Instead, it constructs a synthetic dummy domain (e.g. `urbanmotors.shop`, `novagoods.net`, `apexinteriors.co`).

- Because these domains were procedurally invented and **do not exist on the public internet**, clicking on them in the browser results in `DNS_PROBE_FINISHED_NXDOMAIN` (Cannot be reached).
- This is an intentional simulation feature of the mock dataset to demonstrate broken sites, but creates confusion when clicked in production.

3. Live Interface & API Verification Matrix

MODULE / ENDPOINT	TESTED ACTION / URL	HTTP STATUS	VERIFICATION RESULT & BEHAVIOR
Auth System	POST /api/auth/login	200 OK	Verified session cookie issued, scrypt hash checked, multi-tenant org loaded.
Live Scanner	POST /api/trpc/scanner.scan (linear.app)	200 OK	Live DNS, TTFB, SSL, Cloudflare tech stack, and 3021-day RDAP age calculated.
Discovery Engine	POST /api/trpc/discovery.run	200 OK	Generates candidates, checks DB deduplication, and persists new rows to PostgreSQL.
Lead Table	GET /api/trpc/leads.list	200 OK	Server-side pagination, multi-filter, and sorting functional on Neon PostgreSQL.
CRM Kanban	POST /api/trpc/leads.update	200 OK	Stage transitions (e.g. NEW → RESEARCH) persist and append to activity trail.
Lead Notes	POST /api/trpc/leads.addNote	200 OK	Verified note persistence with author attribution and timestamp.
Campaigns & AI	GET /campaigns/cmp_001	200 OK	Sequence steps load, AI copy generator renders with audit token replacements.
Automation	GET /automation/wf_001	200 OK	React Flow canvas loads with Trigger, Condition, Action, and Wait nodes.

4. Identified Bugs & Recommended Fixes

1. Domain & Company Mismatch (High Priority)

Issue: "Urban Motors" having domain "canva.com" confuses users.

Fix: Update `generateLeads()` so that domain names directly derive from company names (e.g., `urbanmotors.com`) and only use verified live domains from an expanded catalog of 500+ verified businesses.

2. Non-Existent Broken Domains (Medium Priority)

Issue: Clicking synthetic broken domains results in browser NXDOMAIN.

Fix: In the UI, add an indicator badge ("Simulated Offline Domain") or point broken examples to controlled staging testbed URLs (e.g., `broken-test.leadgen.dev`).

3. Scanner Lead Upsert Data Merge (Medium Priority)

Issue: Scanning an existing seeded domain (like `vercel.com`) updates scores but keeps the old mock company name ("Metro Goods").

Fix: Extract the company brand title from the scanned page's `<title>` and `og:site_name` tags to overwrite synthetic names.

4. Phase 2 Real Discovery Integration (Enhancement)

Issue: Discovery relies on synthetic pool generation.

Fix: Connect Google Places API / SerpAPI in `discovery.repo.ts` to fetch real local businesses dynamically based on location and keywords.

Summary Verdict

The application infrastructure on **legends-coral.vercel.app** is fully operational with zero routing or crash errors. The observed repeating domains and broken links are artifacts of the synthetic seed pool in the Discovery and initial database modules, whereas the Website Scanner is genuinely fetching and auditing real live websites.